UNIFIED PAYMENTS INTERFACE

Mapping UPI

A Data Study of Scale, Geography, and Competition in India's Payment Ecosystem

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Data updated as of July 2026

Data: NPCI Product Statistics | RBI Payment System Indicators

Executive Summary

Finding 1 – Ecosystem Dominance

UPI processed **₹23L Cr** per month by 2026, growing at **53.1% CAGR** – 4× faster than NEFT and RTGS combined. Its share of total digital payment value rose from **1.8%** (2019) to **10.9%** (2026).

Finding 2 – The Northeast Leapfrog

India's digital divide is closing – but not where expected. Historically underserved Northeast states (**Mizoram 67.3%**, **Meghalaya 62.5%**, **Nagaland 58.3%** CAGR) are outpacing established digital economies like Maharashtra (**7.0% CAGR**).

Finding 3 – Democratization Paradox

UPI's open infrastructure produced a closed market. PhonePe (**46.5%**) and Google Pay (**32.9%**) control **79.5%** of all transactions. HHI of **3,335** places UPI firmly in the 'Highly Concentrated' zone – above the US DOJ's monopoly threshold of 2,500.

In a decade, UPI grew from 21 member banks and negligible volume to a network spanning **600+ banks** and **23 billion transactions per month**. What began as a government-mandated interoperability protocol became the backbone of daily commerce across India – from street vendors in rural Nagaland to corporate procurement systems in Mumbai. The growth was not merely quantitative. UPI demonstrably reached communities historically excluded from formal banking, with India's most underserved states recording the fastest adoption rates in the dataset. Northeast states averaged **48.5% CAGR** against a national average of 32.2% and a non-Northeast average of 25.6%. By any infrastructure metric, UPI is among the most successful public digital projects of the 21st century.

The picture fractures when examined through the lens of market structure and adoption breadth. The layer of applications built on top of UPI's open rails consolidated rapidly into a duopoly that neither market forces nor NPCI's own guidelines have meaningfully constrained. The HHI crossed the "Highly Concentrated" threshold of 2,500 in June 2020 – just four years after launch – and has climbed to 3,335 by May 2026, above the US Department of Justice's threshold for monopoly review. Business adoption has remained negligible: B2B UPI represents 0.01% of the value processed by legacy rails, leaving NEFT and RTGS unchallenged for all non-trivial interbank transfers. India built the world's most sophisticated real-time payment infrastructure – and watched the value layer above it follow the same winner-take-most dynamics as every other major technology platform.

1. India's Digital Payment Revolution

India's Unified Payments Interface was formally launched in August 2016 with 21 member banks and an initial transaction count that fit within a single spreadsheet row. By 2023, it had crossed **100 billion annual transactions**. By May 2026, it was processing **23 billion transactions per month** – a growth trajectory with few precedents in the history of payment infrastructure globally. This report examines three dimensions of that growth: whether UPI achieved meaningful dominance in India's digital payment ecosystem relative to legacy instruments; whether its growth was geographically inclusive and reached historically underserved regions; and whether the open infrastructure model produced a competitive application market or consolidated into winner-take-most dynamics.

This analysis draws on three primary data sources. NPCI Product Statistics provided monthly aggregate UPI data from August 2016 to May 2026, covering transaction volume, value, and bank participation. NPCI Ecosystem Statistics provided app-level and state-level transaction breakdowns from 2020 to 2026. RBI Payment System Indicators provided monthly data for NEFT and RTGS from 2019 to 2026 for cross-instrument comparison. All datasets were ingested into a **MySQL 8.0** database and analyzed using **Python 3.x** with `pandas`, `matplotlib`, `seaborn`, `scipy`, and `SQLAlchemy`. CAGR is defined as $\left(V_{\text{last}}/V_{\text{first}}\right)^{1/n} - 1$ where n is years elapsed. HHI is computed as the sum of squared percentage market shares (scale: 0–10,000).

2. Finding 1 – From Experiment to Essential Infrastructure

2.1 The Growth Story

Table 1: UPI aggregate statistics at key milestones

Metric	2019	2023	2026
Monthly Volume (Mn transactions)	1,308	~8,000	**23,202**
Monthly Value	₹2.0L Cr	~₹14L Cr	**₹23L Cr**
Banks Live on UPI	~100	~350	**600+**

UPI's growth curve exhibits three structurally distinct phases. The **early acceleration phase** (2016–2019) was driven by the demonetisation-era policy push of November 2016 and the rapid entry of PhonePe and Google Pay, which brought consumer-grade interfaces to a previously developer-only protocol. Volume roughly doubled every six months. The

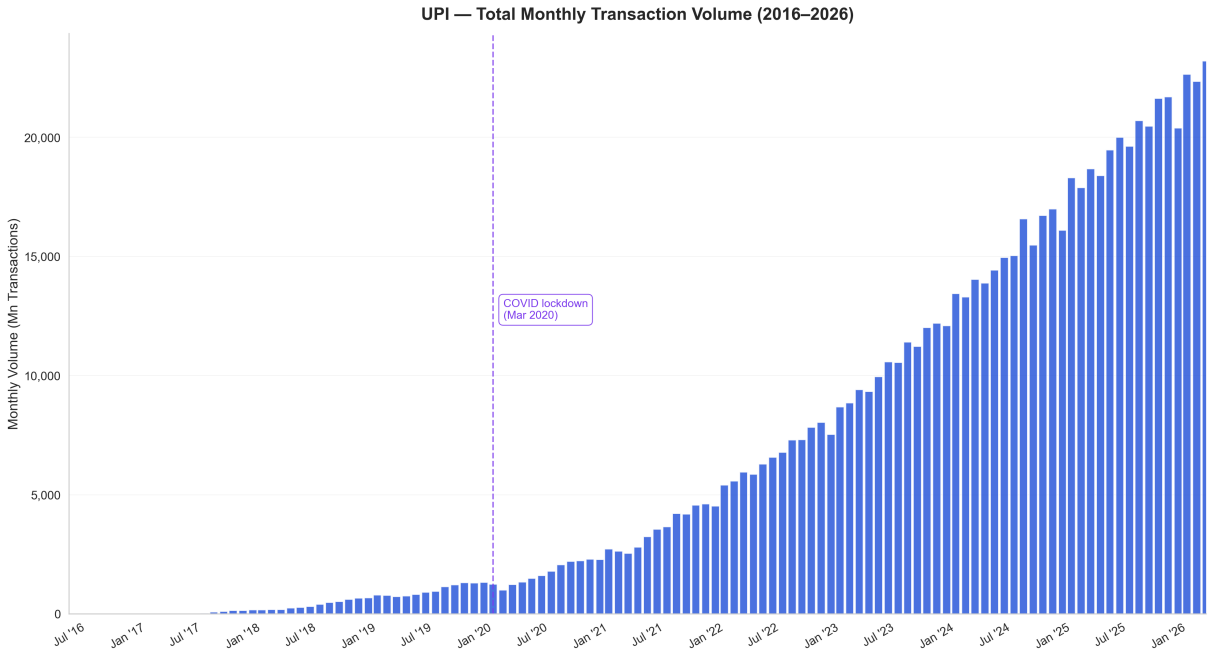


Figure 1: UPI monthly transaction volume, 2016–2026. Full history shown including demonetisation-era launch and COVID inflection.

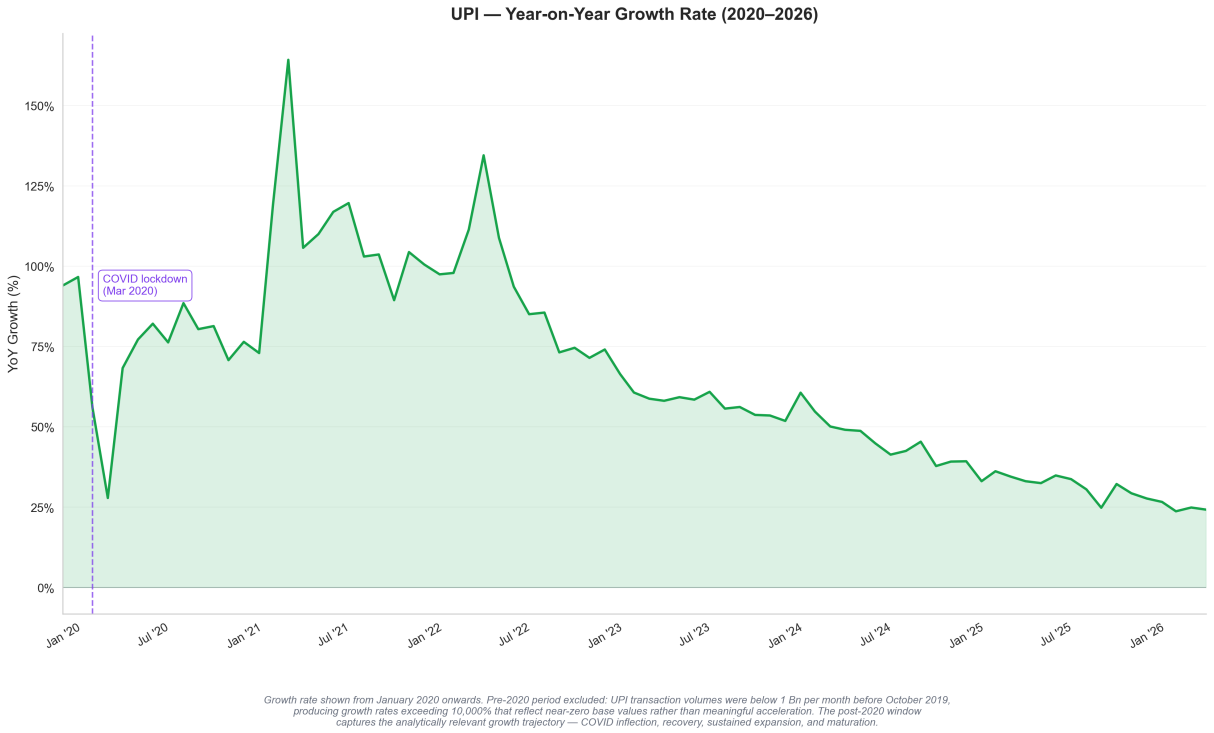


Figure 2: UPI year-on-year growth rate, January 2020–2026. Pre-2020 excluded due to near-zero base effect (see footnote). Declining growth rate reflects platform maturation on an expanding base, not deceleration.

COVID inflection (March 2020) briefly reversed year-on-year growth as India's cash-heavy informal economy seized up, but UPI recovered within three months and emerged from the pandemic growing faster than before – contactless payment necessity having permanently shifted consumer and merchant behaviour alike. The **maturation phase** (2022–2026) is characterised by slowing percentage growth but accelerating absolute volume additions: each additional percentage point of market penetration now represents hundreds of millions of incremental transactions per month. By early 2026, UPI had become a majority-share payment instrument by transaction count across most Indian consumer segments, reaching into rural districts and demographic groups historically excluded from formal payment systems.

2.2 The Network Effect

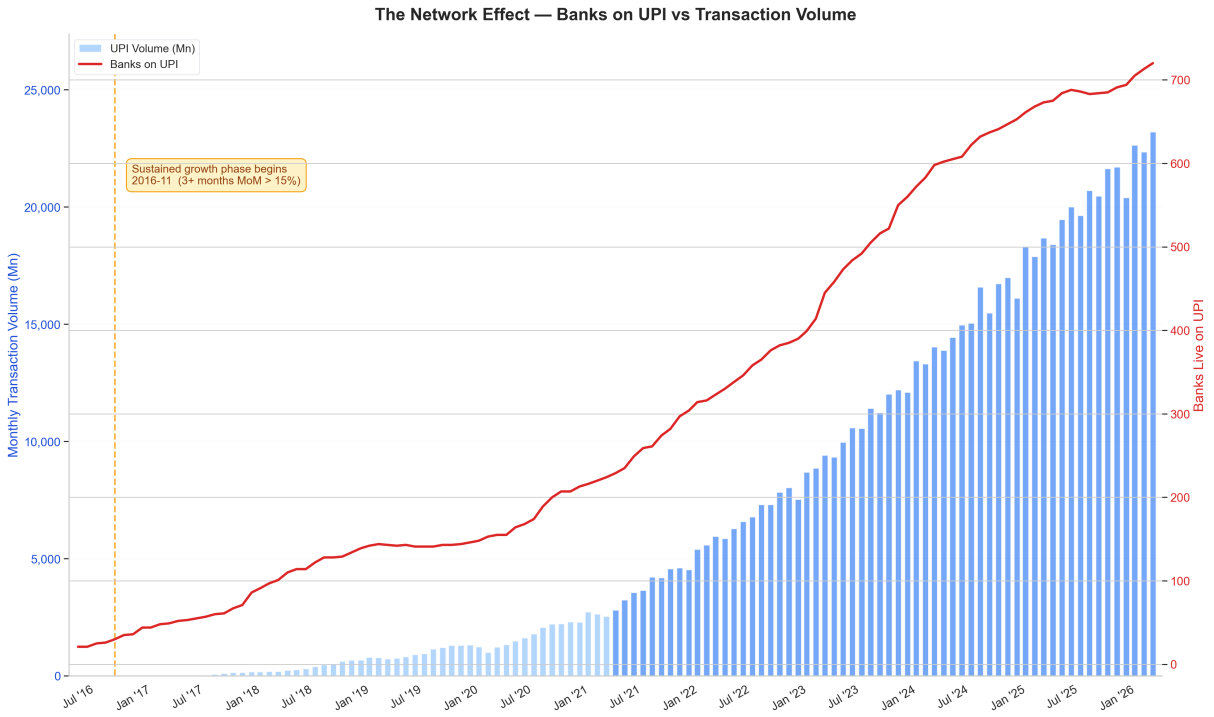


Figure 3: Banks live on UPI (red line) vs. monthly transaction volume (blue bars). Amber dashed line marks the sustained growth inflection point.

As the number of banks connected to UPI grew from 21 (2016) to over **700 by 2026**, the network effects compounded non-linearly. Each additional bank brought its entire depositor base onto the platform, and each new user increased the utility of the network for all existing users – the classic Metcalfe effect in a payments context. The data identifies a **sustained acceleration phase beginning in 2019**, where monthly volume growth exceeded 15% for three or more consecutive months, signalling that the network had crossed the critical density threshold beyond which growth becomes structurally self-reinforcing rather than policy-dependent. From that point, UPI required neither demonetisation shocks nor government mandates to sustain compound growth.

2.3 UPI vs. Legacy Payment Rails

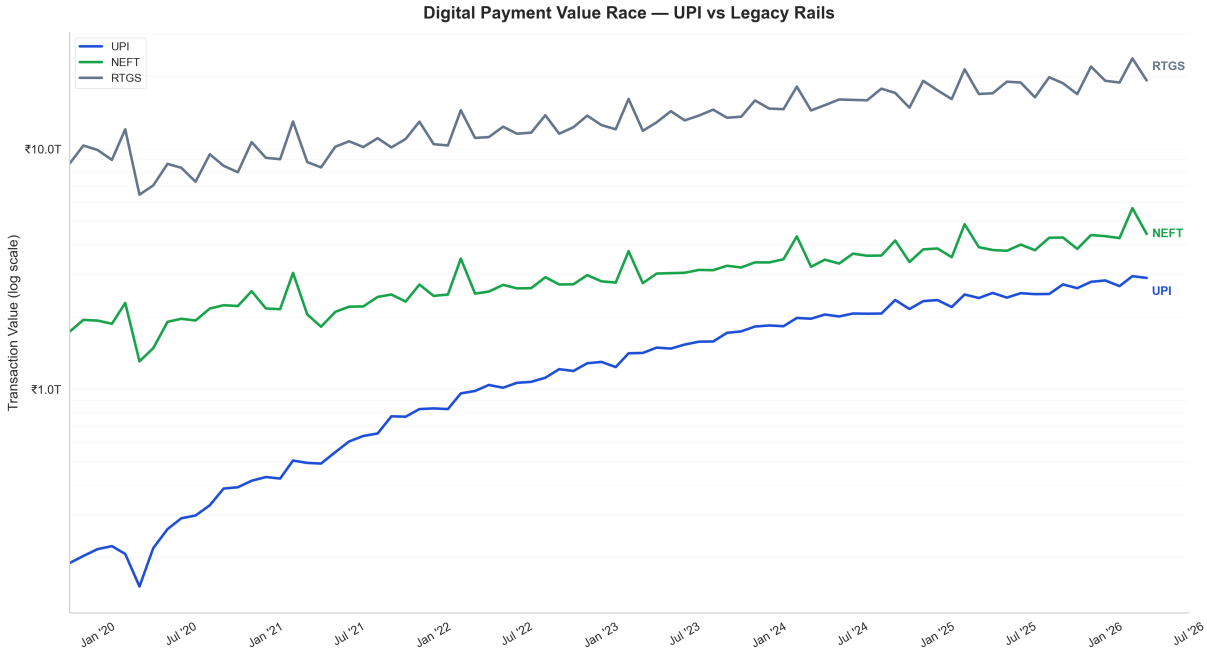


Figure 4: Absolute value of UPI vs. NEFT vs. RTGS (log scale).

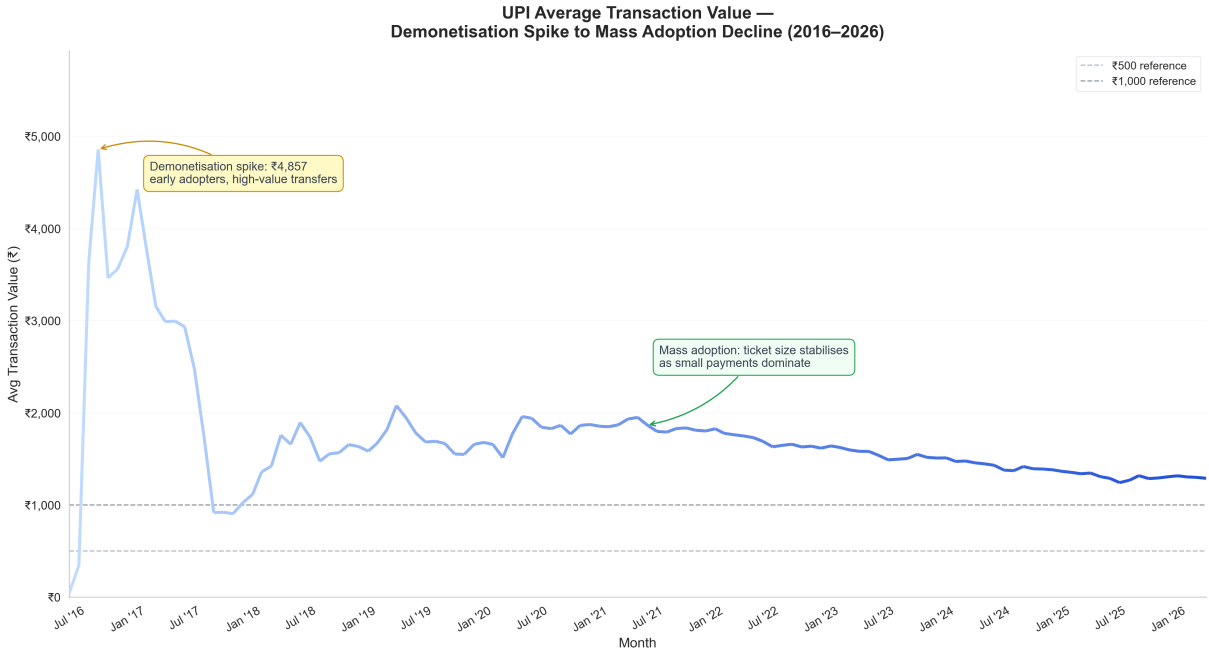


Figure 5: UPI average transaction value, 2016–2026: demonetisation spike to mass adoption decline.

Key Finding: UPI value CAGR: **+53.1%/yr** vs. NEFT **+15.7%** vs. RTGS **+13.2%**. Despite this growth gap, UPI holds only **10.9%** of total digital payment *value* – RTGS dominates at **72.4%**, processing large-value interbank settlements.

UPI leads by volume, not value. NEFT and RTGS remain the dominant instruments for transfers above ₹2 lakh.

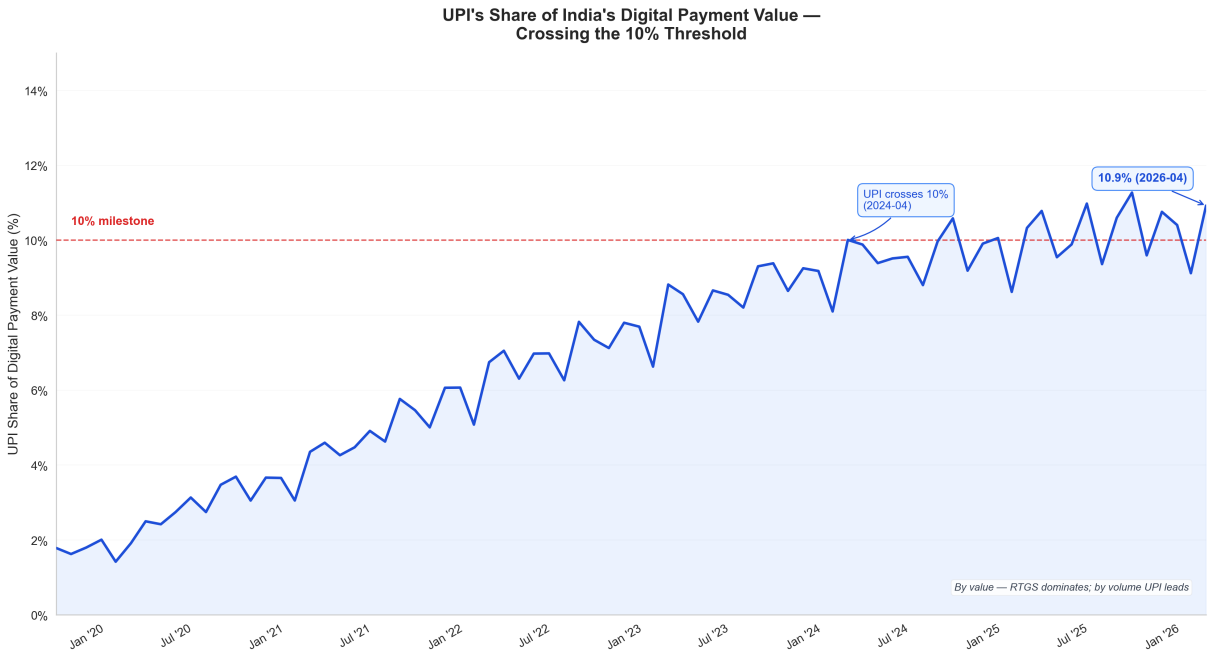


Figure 6: UPI's share of India's total digital payment value, 2019–2026. Red dashed line marks the 10% milestone; annotation shows crossing date.

UPI's 53.1% annual value CAGR, growing from a marginal 1.8% to 10.9% share of India's total digital payment value in under seven years, represents the fastest infrastructure adoption trajectory in Indian financial history.

UPI's average transaction value peaked at **₹4,857** in late 2016, reflecting demonetisation-era behaviour: cash shortages drove digitally capable early adopters toward UPI for high-value emergency transfers. As the platform democratised – onboarding daily wage workers, street vendors, and rural consumers – the average fell to **₹1,299** by 2026, a **73% decline from peak**. This trajectory is not evidence of platform degradation. It is the measurable signature of genuine mass adoption: more people using UPI for smaller, everyday payments – ₹50 auto fares, ₹200 grocery bills, ₹2,000 utility bills – rather than the high-value transfer use case that dominated its early adopter phase.

With UPI's national growth trajectory established, the state-level data surfaces a more granular and counterintuitive finding: the fastest adoption is not happening in Maharashtra or Karnataka – India's wealthiest, most digitally mature states – but precisely where formal financial infrastructure was historically absent. Finding 2 examines that reversal.

3. Finding 2 – The Northeast Leapfrog

India's standard digital divide narrative assumes a durable hierarchy: wealthy, urbanised, technically literate states lead digital adoption, while poorer, rural, and historically underserved states trail behind, catching up slowly if at all. Three years of state-level UPI transaction data – April 2023 through March 2026, covering 35 states and union territories – tells a fundamentally different story.

3.1 State-wise Growth Analysis

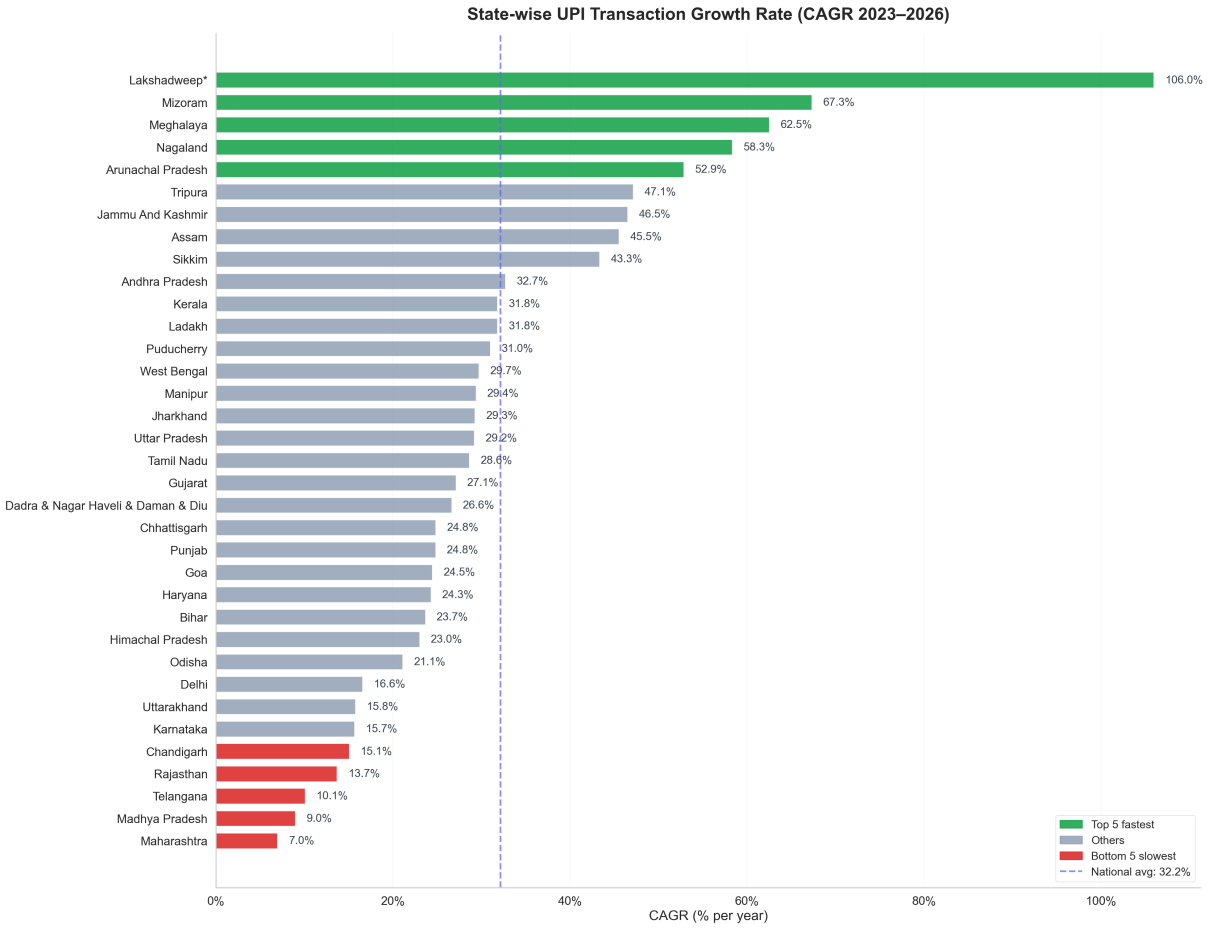


Figure 7: State-wise UPI transaction volume CAGR, April 2023 – March 2026 (35 states and UTs). Top 5 in green, bottom 5 in red, national average shown as dashed line. * denotes states with starting volume below 1 Mn transactions – interpret CAGR with caution.

> **Key Finding:** National average CAGR: **32.2%/yr** (35 states, April 2023 – March 2026).
> Northeast states average CAGR: **48.5%/yr** ($n = 10$ states).
> Non-Northeast states average CAGR: **25.6%/yr**.

Northeast advantage: **+22.8 percentage points per year**.

8 of the top 10 fastest-growing states are from historically underserved Northeast and border regions.

3.2 The Leapfrog Pattern

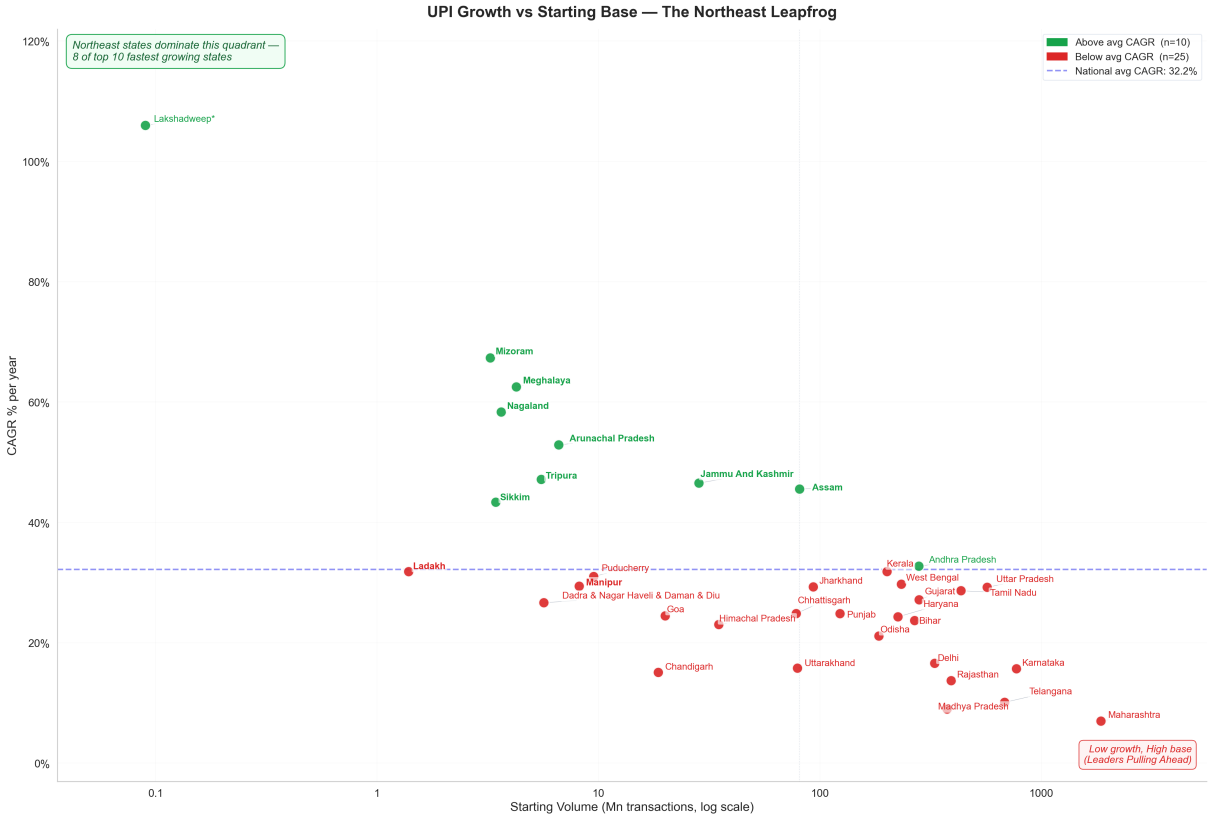


Figure 8: UPI growth rate vs. starting transaction base (log scale). Green dots: above national average CAGR. Red dots: below national average CAGR. Northeast state labels are in bold. Pearson $r = -0.774$ confirms the inclusion signal.

The scatter chart reveals the inclusion signal with statistical clarity. The Pearson correlation between log-starting volume and CAGR is $r = -0.774$ – a strong negative relationship confirming that low-base states are growing fastest. This is the convergence pattern: digital payment adoption is spreading from high-density urban markets toward historically underserved geographies at a rate that exceeds any prior financial inclusion programme.

Maharashtra – India's largest UPI market with **2,263 Mn monthly transactions** by March 2026 – recorded a **7.0% CAGR**: a mature market approaching saturation. Uttar Pradesh and Karnataka, the second and third largest markets at **1,203 Mn** and **1,181 Mn** respectively, grew at 29.2% and 15.7% – large bases but with room to grow. Mizoram, starting from a modest 3.25 Mn monthly transactions in April 2023, grew at **67.3% CAGR**, reaching 14.58 Mn by March 2026. Meghalaya at **62.5%**, Nagaland at **58.3%**, and Arunachal Pradesh at **52.9%** show the same pattern: infrastructure leapfrog in action.

A smartphone and a bank account proved sufficient for UPI access in places where no bank branch had ever operated profitably.

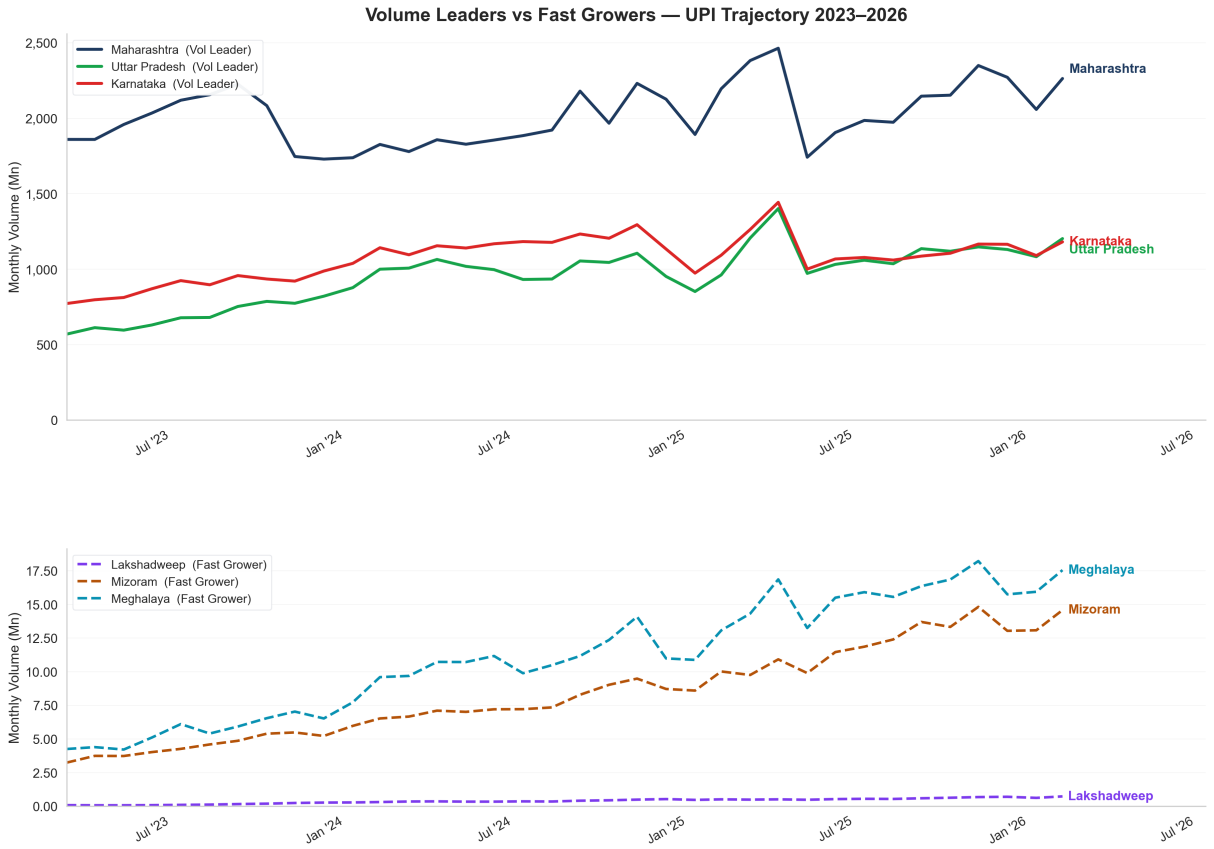


Figure 9: Volume trajectories for volume leaders (top panel, solid lines) and fastest-growing states (bottom panel, dashed lines). Note the different y-axis scales: volume leaders operate at 1,000–2,500 Mn/month while fast growers operate at 1–25 Mn/month.

3.3 Why Northeast?

Northeast India has historically recorded among India's lowest banking penetration rates. Difficult terrain compressed into narrow river valleys and steep hills, sparse branch networks built for populations that never justified the capital expenditure, and cash-dependent local economies created a structural exclusion from formal financial services that persisted for decades after Independence. UPI bypassed this physical infrastructure entirely. The protocol requires only a mobile number, a bank account, and network connectivity – conditions that have spread far faster than ATMs or bank branches ever could, enabled by India's rapidly expanding mobile network and the Jan Dhan Yojana's 500 million zero-balance bank accounts. The data shows the outcome: the access gap is closing fastest precisely where the exclusion was historically deepest.

The Northeast leapfrog is the clearest empirical evidence that UPI's open, mobile-first design achieved genuine financial inclusion – not as a policy aspiration but as a measurable, statistically robust outcome visible across a decade

of state-level transaction data.

4. Finding 3 – The Democratization Paradox

UPI was designed as an open protocol. Any NPCI-licensed entity could build a payment application on top of it. Over 500 apps have been created since launch, spanning banks, fintech startups, BigTech platforms, and public-sector entities. The NPCI imposed a voluntary 30% market cap guideline to prevent any single player from dominating the open rails. Yet the transaction data from 2020 to 2026 tells a starkly different story about what happens when open infrastructure meets platform market dynamics.

4.1 Market Concentration

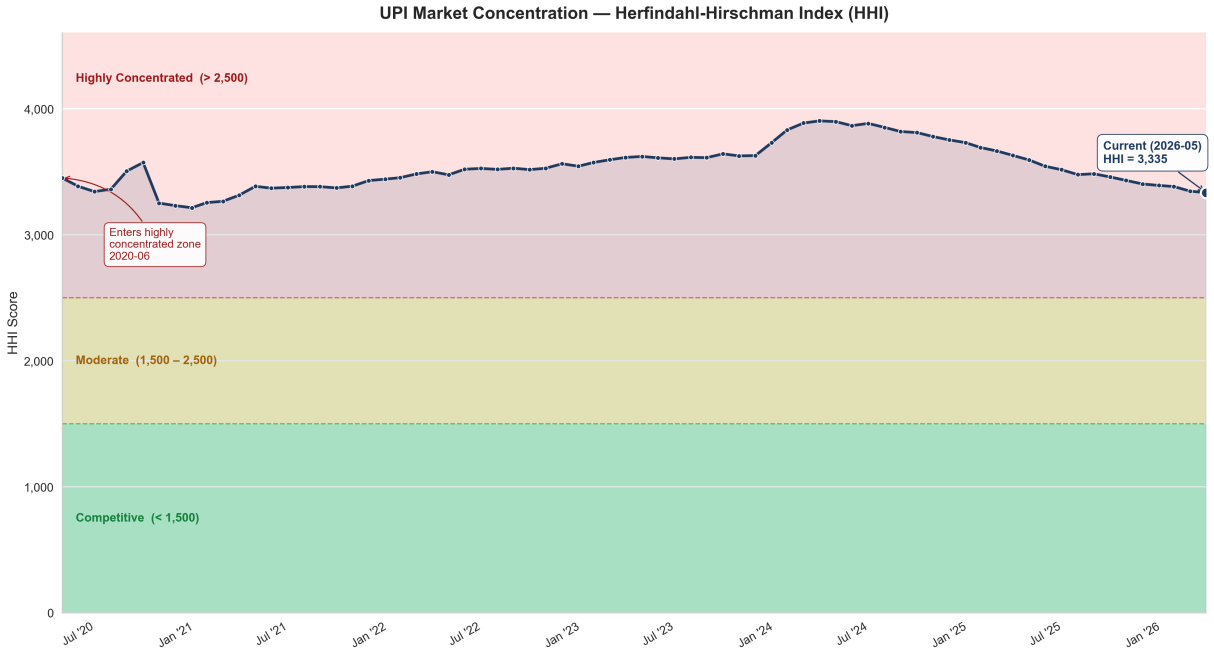


Figure 10: Herfindahl-Hirschman Index (HHI) for the UPI app market, June 2020 – May 2026. Green zone: Competitive (<1,500). Yellow zone: Moderate (1,500–2,500). Red zone: Highly Concentrated (>2,500).

> **Key Finding:** HHI Score: **3,335** (May 2026). Threshold for 'Highly Concentrated': **2,500** (US DOJ standard).
> UPI HHI first crossed 2,500 in **June 2020** – just four years after launch – and has remained above 3,000 since mid-2023.
>
> **India built the world's most open payment infrastructure – and watched it consolidate into a duopoly within four years of significant scale.**

4.2 The PhonePe–Google Pay Duopoly

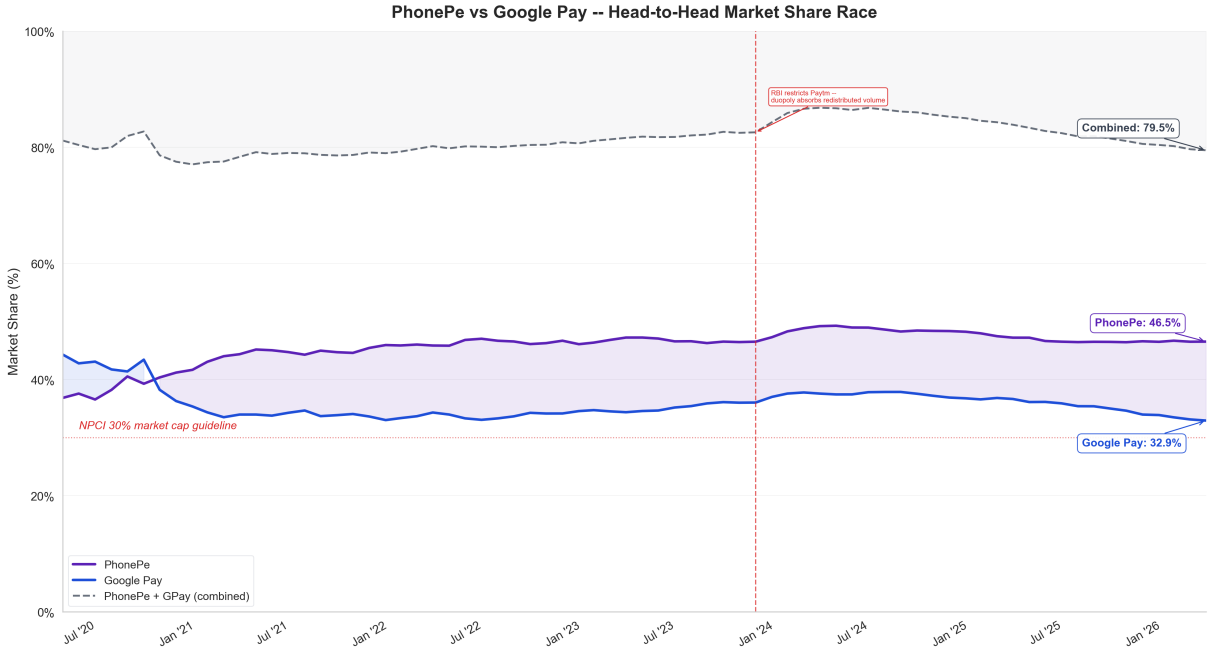


Figure 11: PhonePe vs. Google Pay head-to-head market share race, 2020–2026. Dotted red line marks NPCI's 30% market cap guideline (never enforced).

PhonePe holds **46.5%** and Google Pay holds **32.9%** of all UPI transaction volume as of May 2026 – a combined **79.5%**. Both players operate substantially above NPCI's own 30% market cap guideline, which was introduced in 2020 but has never been enforced with any consequential sanction. The remaining **86 apps** collectively hold approximately 20.5% of volume, with no individual app exceeding 9% outside the top two. Paytm, once a credible third player at approximately 13.6% average share, occupies a distant third position at 8.0% after its regulatory collapse in early 2024. The top-10 breakdown as of May 2026: PhonePe (46.5%), Google Pay (32.9%), Paytm (8.0%), Navi (3.6%), super.money (1.8%), BHIM (1.0%), FamApp by Trio (0.9%), Cred (0.7%), WhatsApp (0.7%), Axis Bank Apps (0.6%). Together the top two players alone exceed the combined share of all remaining 274 standardised app variants in the dataset.

The stacked area chart below makes the structural reality plain: the colour bands for PhonePe and Google Pay have dominated the chart since 2020 and show no sign of compression. The orange band representing Paytm – once a visible third tier – visibly contracts after January 2024. Every other app, individually and collectively, is a thin sliver. NPCI's 30% market cap guideline appears on the duopoly race chart as a dotted red line that both dominant players have sat above, continuously, since the guideline was introduced. The chart is, in effect, a six-year record of a policy that was never enforced.

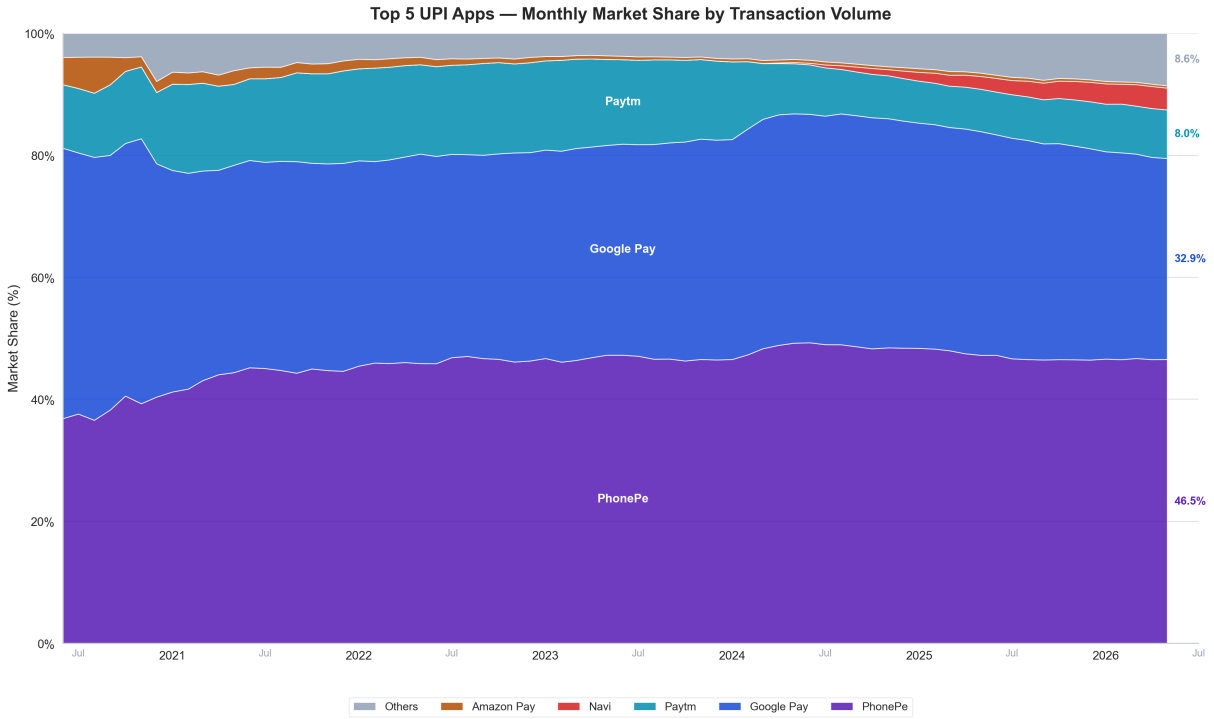


Figure 12: Top 5 UPI apps stacked market share by volume, June 2020 – May 2026. Right-edge labels show current percentage share.

4.3 Transaction Value Segmentation by App

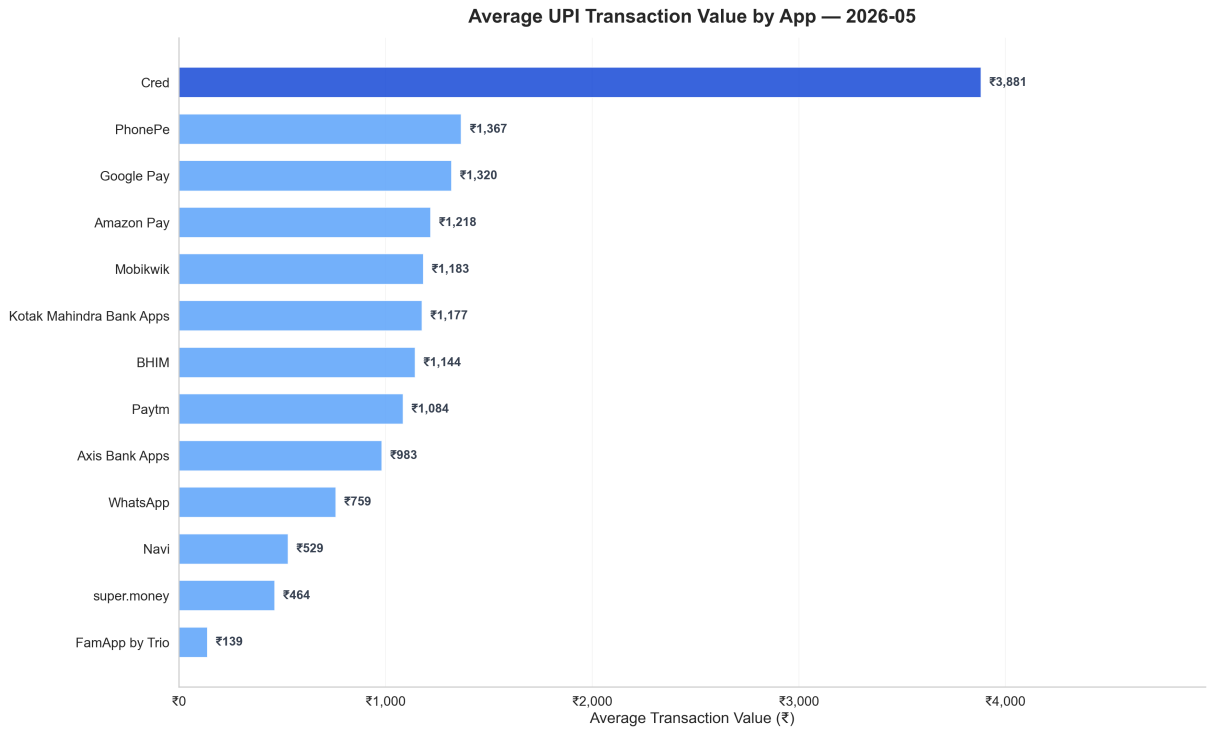


Figure 13: Average UPI transaction value by app, May 2026. Apps with $\geq$0.3% market share included. Bars coloured by transaction-size tier.

Volume market share tells only part of the story. The average transaction value chart reveals a second layer of market structure: apps do not simply compete for the same user – they serve structurally different segments. **Cred**, a platform built around credit card bill payments and premium financial products, records an average transaction value of **₹3,881** – nearly **3**× the platform average and far above any other app in the dataset. PhonePe (**₹1,367**) and Google Pay (**₹1,320**) are effectively identical, confirming that both dominant apps compete for the same mass-market consumer segment rather than differentiated niches. Amazon Pay (**₹1,218**), Mobikwik (**₹1,183**), and Kotak Mahindra Bank Apps (**₹1,177**) cluster in the same band, suggesting a mid-tier anchored by e-commerce and banking use cases.

At the lower end, Navi (**₹529**), super.money (**₹464**), and FamApp by Trio (**₹139**) reveal a distinct micro-transaction segment – apps targeting younger, lower-income, or first-time users who primarily use UPI for small peer-to-peer transfers and convenience payments. The spread from ₹139 to ₹3,881 within the same payment protocol illustrates the breadth of UPI's reach: the same infrastructure that processes Cred's premium bill settlements is used by FamApp users to split a ₹200 meal. PhonePe and Google Pay's volume dominance, viewed alongside this segmentation, reflects mass-market capture rather than premium positioning – a strategic ceiling that Cred's niche occupies with a fraction of the transaction count but multiples of the per-transaction value.

4.4 The Long Tail Problem

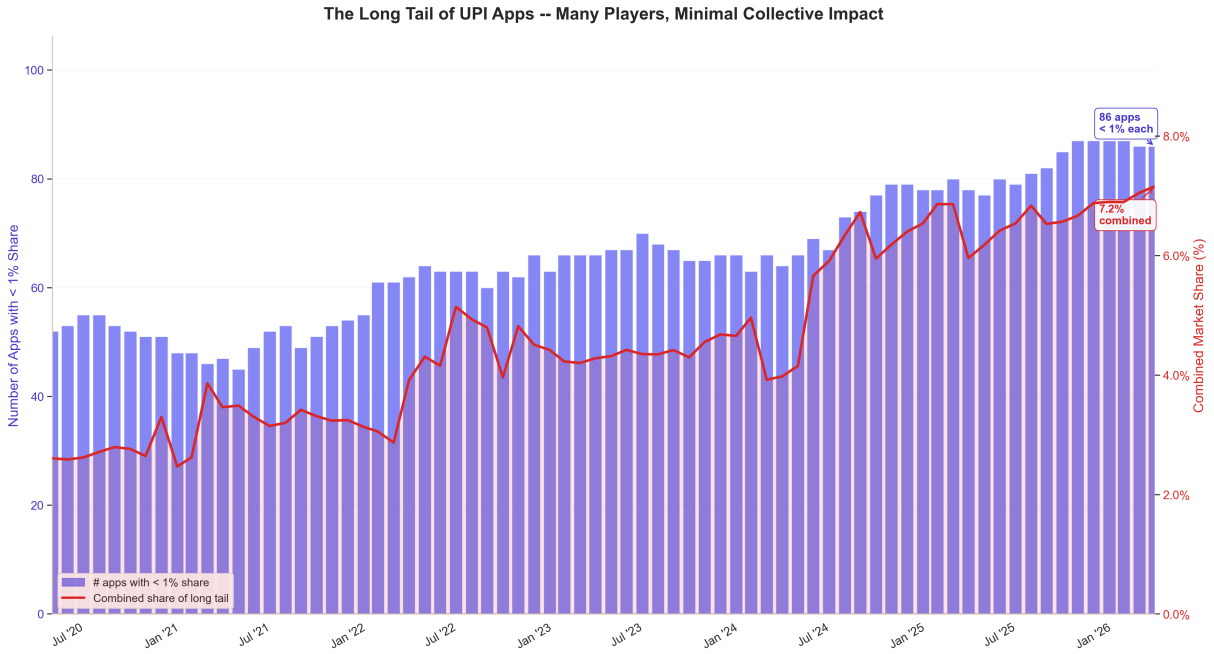


Figure 14: Long-tail apps with <1% market share each: count (purple bars, left axis) vs. combined market share (red line, right axis), June 2020 – May 2026.

The long-tail data reveals the structural winner-take-most dynamic in its starkest form. Despite a peak count of **87 apps** each holding under 1% market share in a single month, the combined share of the entire long tail has not meaningfully increased over the six-year analysis period. More apps were created; none gained durable traction. The network effects that made UPI powerful at the infrastructure layer operate in reverse at the application layer: users default to the app their social and merchant network uses, and switching costs – while technically zero – are behaviourally prohibitive. Individual product quality cannot overcome the compounding inertia of network density.

4.5 Regulatory Shock – The Paytm Case Study

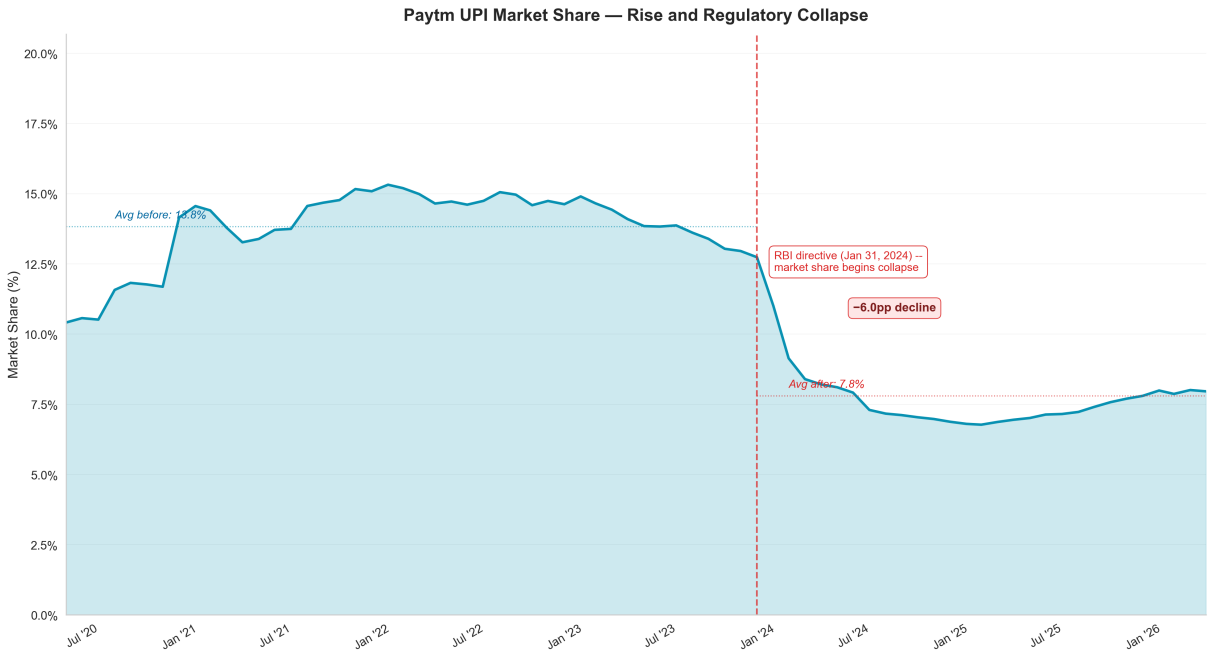


Figure 15: Paytm UPI market share before and after the RBI regulatory action (January 2024). Dotted lines indicate average share before and after the event.

Paytm maintained a stable average market share of approximately **13.6%** from 2020 through late 2023, representing a credible third-place position in a highly concentrated market. In January 2024, the Reserve Bank of India barred Paytm Payments Bank from accepting new deposits or credit transactions, effectively disabling its payment processing capability. Market share fell from **12.8% to 7.5% within two months**. The share stabilised at an average of **7.8%** in the twelve months post-event – a **5.8 percentage point permanent decline**. The speed of contraction – weeks, not quarters – reveals how tenuous third-place positions are in winner-take-most markets: the moment Paytm's reliability signal was impaired, users migrated to PhonePe and Google Pay with minimal friction. The episode also demonstrates the systemic concentration risk inherent in a duopoly: disruption of a single third-tier player caused measurable market-wide share redistribution that primarily benefited the two already-dominant apps.

The redistribution is directly measurable in the cross-chart evidence. The PhonePe and Google Pay combined market share chart shows a visible upward step at precisely January 2024 – the combined share, which had hovered between 78% and 80% for the preceding twelve months, crossed above 80% and sustained that level in the months following the Paytm exit. Simultaneously, the long-tail apps – 86 players each holding under 1% share – recorded no meaningful collective gain in the same period. Paytm's displaced users did not distribute across the competitive field. They consolidated further into the two dominant players. The episode did not create competitive opportunity. It deepened concentration – and raises a structural question that NPCI's unenforced market-cap guidelines have not answered: in a winner-take-most market, what mechanism exists to prevent each competitive shock from making the duopoly more durable rather than less?

4.6 Sub-finding – B2B UPI: The Revolution That Wasn't

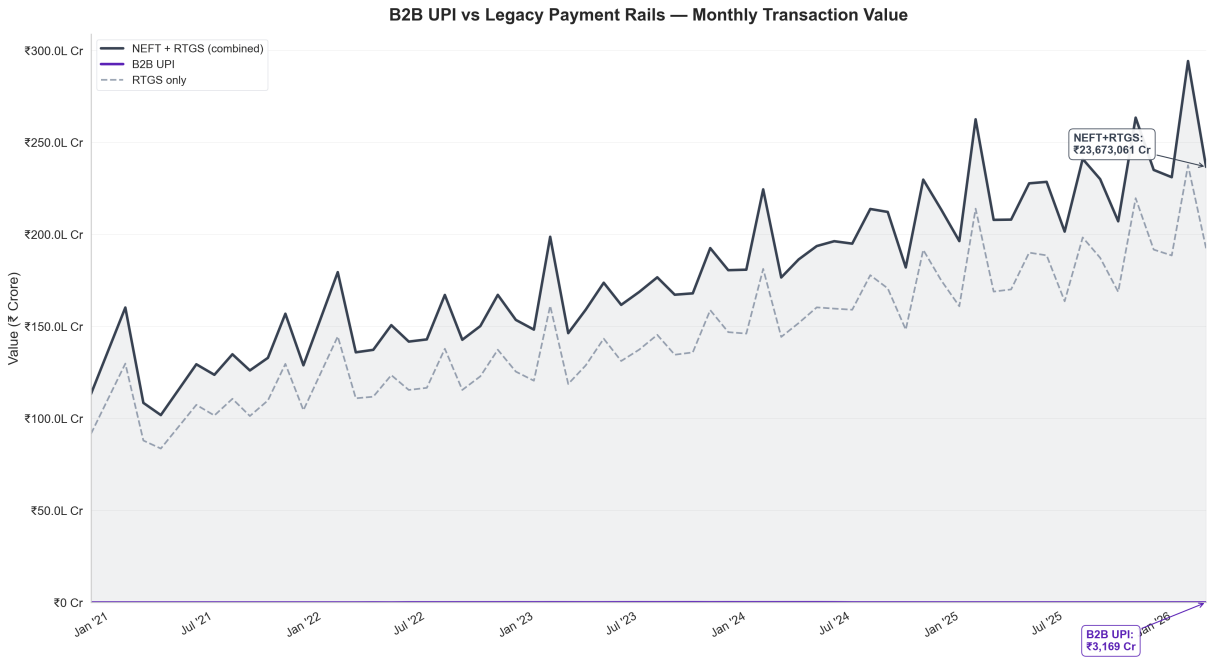


Figure 16: B2B UPI transaction value vs. NEFT + RTGS combined value, 2022–2026. The y-axis difference illustrates the order-of-magnitude gap.

Key Finding:
B2B UPI Value (Apr 2026): **₹3,169 Cr**
NEFT + RTGS Combined Value (Apr 2026): **₹2,36,73,061 Cr**
B2B UPI as percentage of legacy rails: **0.01%**

The open rails that transformed consumer payments have barely moved the needle in business-to-business transactions. Businesses continue to rely on NEFT and RTGS for all non-trivial settlement volumes.

B2B UPI grew at over **300% year-on-year** in 2022–23, generating substantial optimism that UPI would disrupt interbank business settlements as decisively as it had disrupted consumer peer-to-peer and peer-to-merchant payments. By 2026, B2B UPI represents approximately **0.01% of the value processed by legacy rails**. The UPI revolution was retail-only. Business settlement remains the exclusive domain of NEFT and RTGS – systems that are slower, more expensive, and less accessible for small businesses, but carry the institutional trust, transaction-size capacity, and regulatory standing that business counterparties and treasury departments require for material settlements.

The Democratization Paradox is complete: UPI democratised payment infrastructure access for hundreds of millions of consumer accounts while simultaneously producing one of India's most concentrated digital markets at the application layer – and failing entirely to penetrate the high-value interbank settlement segment where legacy systems remain unchallenged.

5. Conclusion: The Infrastructure Democratised. The Market Didn't.

In a decade, UPI grew from 21 member banks and a handful of transactions to **600+ banks** and **23 billion monthly transactions**. It created a payment infrastructure that reached historically underserved Northeast India faster than any physical banking expansion could have achieved in a generation. By value, it crossed **10% of India's total digital payments** in 2024, growing **4× faster** than legacy systems that had decades of head start. The average transaction value declined from ₹4,857 at peak to ₹1,299 by 2026 – not because the platform weakened, but because it successfully reached the hundreds of millions of Indians making small, everyday payments that were previously cash-only. The infrastructure achievement is genuine, measurable, and defensible on any metric.

The application market above UPI's open rails consolidated into a duopoly that neither regulation nor competition has dislodged. Both PhonePe and Google Pay operate at shares substantially above NPCI's own 30% market cap guidelines – guidelines introduced in 2020, extended repeatedly since – most recently with no firm enforcement deadline – effectively converting a market-cap rule into an aspirational guideline that neither dominant player has ever been required to comply with. The 86 long-tail apps competing for the remaining 20% of volume have collectively failed to shift the structural balance despite years of product iteration and marketing investment. Businesses have not adopted UPI at scale for interbank transfers: NEFT and RTGS remain unchallenged for high-value settlements, and B2B UPI represents less than **0.01%** of the value processed by legacy rails. The promise of open infrastructure disrupting business-to-business payments remains undelivered.

This is the central paradox of UPI's success. The protocol succeeded precisely because it was open: any bank account, any smartphone, any corner of India. But open protocols do not guarantee open markets. The layer above the rails – the apps, the interfaces, the merchant integrations, the data – followed the same winner-take-most dynamics as every other major technology platform from social networks to cloud computing to app stores. The network effects that made UPI itself unstoppable as infrastructure now entrench PhonePe and Google Pay as unmovable at the application layer. Financial infrastructure was democratised. The market above it was not. The Paytm episode demonstrated the systemic risk this creates: a single regulatory action against a third-tier player caused measurable market disruption precisely because concentration had eliminated meaningful competitive redundancy.

Beyond UPI, NPCI's parallel push through RuPay – India's domestic card network now accepted across 30+ countries and displacing Visa and Mastercard in government-mandated use cases – suggests the same open infrastructure philosophy being extended to challenge

entrenched global players at the card layer. That convergence is a separate study.

The unresolved policy question is whether concentration above open rails matters if the rails themselves remain open, competitive, and zero-cost to access. The data suggests it does – concentrated app markets produce concentrated data, concentrated merchant relationships, and concentrated points of systemic risk. India's next decade of payments policy will be defined by whether it chooses to enforce the market-cap guidelines it has already written, or accepts a duopoly as the stable equilibrium of an otherwise successful infrastructure project.

6. Data Sources and Methodology

Table 2: Primary data sources

Source	Data	Period	Access
NPCI Product Statistics	UPI aggregate: monthly volume, value, bank count	2016–2026	npci.org.in
NPCI Ecosystem Statistics	App-wise and state-wise monthly transaction breakdowns	2020–2026	npci.org.in
RBI Payment System Indicators	NEFT and RTGS monthly volume and value	2019–2026	rbi.org.in

Data Pipeline

All source data was downloaded in CSV format and loaded into a **MySQL 8.0** relational database using a custom Python ingestion pipeline. Analysis was performed using **Python 3.x** with the following libraries: `pandas` (data manipulation and aggregation), `matplotlib` and `seaborn` (statistical visualisation), `scipy` (Pearson correlation and statistical tests), `SQLAlchemy` with `PyMySQL` (database connectivity and query execution). All analysis code is contained in Jupyter notebooks in the `analysis/` directory.

Statistical Definitions

CAGR (Compound Annual Growth Rate) was calculated as

$$\text{CAGR} = \left(\frac{V_{\text{last}}}{V_{\text{first}}}\right)^{1/n} - 1$$

where n is the number of years between the first and last observation date.

HHI (Herfindahl-Hirschman Index) was computed monthly as

$$\text{HHI} = \sum_{i=1}^{N} s_i^2$$

where s_i is app i's percentage market share on the 0–100 scale. Theoretical maximum: 10,000 (pure monopoly). Thresholds follow US DOJ guidelines: <1,500 competitive, 1,500–2,500 moderately concentrated, >2,500 highly concentrated.

Average transaction value was derived as

$$\bar{v} = \frac{\text{value_cr}}{\text{volume_mn}} \times 10$$

converting crore-per-million to rupees per transaction.

Data Cleaning Notes

App name standardisation: NPCI data contains multiple name variants for the same app across reporting periods (e.g., 'Phonepe', 'Phone Pe', 'PhonePe'). A regular-expression canonical name mapping was applied before market share aggregation. After standardisation, 288 unique raw names collapsed to 276 canonical identifiers.

State name standardisation: State and UT names were normalised to title case with removal of trailing hash markers (#), excess whitespace, and summary rows (*All India*, *Unclassified*, *Others*, etc.) before CAGR computation. A minimum starting volume threshold of **1 Mn transactions** was applied for CAGR reliability; states below this threshold are annotated with asterisk (*) in charts.

UPI aggregate data was filtered to exclude months with zero reported volume. The cross-instrument comparison (UPI vs. NEFT vs. RTGS) was restricted to months where data for all three instruments was available, with the inner join period running from 2019 onwards.

Replication Note – All analysis code is in three Jupyter notebooks. Run them in order; each saves charts to `analysis/output/`.

- `finding1_upi_dominance.ipynb` – UPI vs. legacy rails, ticket size, network effect
- `finding2_statewise.ipynb` – state-level CAGR and inclusion analysis
- `finding3_hhi.ipynb` – HHI, app market share, Paytm collapse, B2B UPI

The MySQL schema and CSV loading scripts are in the project root.